

SIMATS ENGINEERING

Department of Computer Science and Engineering



CSA15 Cloud Computing and Big Data Analytics

CO2 AT2 - VM Configuration and Security Evidence Workbook

Guided practical workbook for Azure VM, local VM, Docker container, security checks, recovery evidence and GitHub submission

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT2 - VM Configuration and Security Evidence
Submission Type	Individual digital workbook based on the same capstone title used in CO1 and CO2 AT1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook, evidence screenshots and GitHub repository link through Google Classroom

How This Workbook Works

- Use the same capstone title carried from CO1 and CO2 AT1.
- Read each procedure block, perform the action, verify the checkpoint and then fill the response table.
- Use Azure first when account activation is available. Use the local VM path when Azure is blocked. Use Docker for container comparison. Use simulation only when practical deployment is blocked.
- Screenshots must show the student's actual configuration or verified simulation work. Hide credentials, keys, tokens and sensitive account details.
- Upload the completed workbook and evidence to GitHub, then submit the GitHub repository link through Google Classroom.

Virtualization Tracks for CO2 AT2

Azure is the preferred cloud path. Local VM and container tracks are included for hands-on resilience.

Track A - Azure Linux VM

Preferred cloud evidence: subscription, VM, SSH, NGINX/test service, firewall, shutdown

Track B - Local VM

VirtualBox or VMware with Tiny Core, Alpine, Debian netinst or Ubuntu Server

Track C - Docker Container

Run a lightweight service, map a port, inspect logs, compare with VM

Track D - Simulation

Used only if deployment is blocked; must include technical configuration and security evidence

Required common output: completed workbook, screenshots, command outputs, security choices, README.md, GitHub repository link and GCR submission.

Technical Orientation

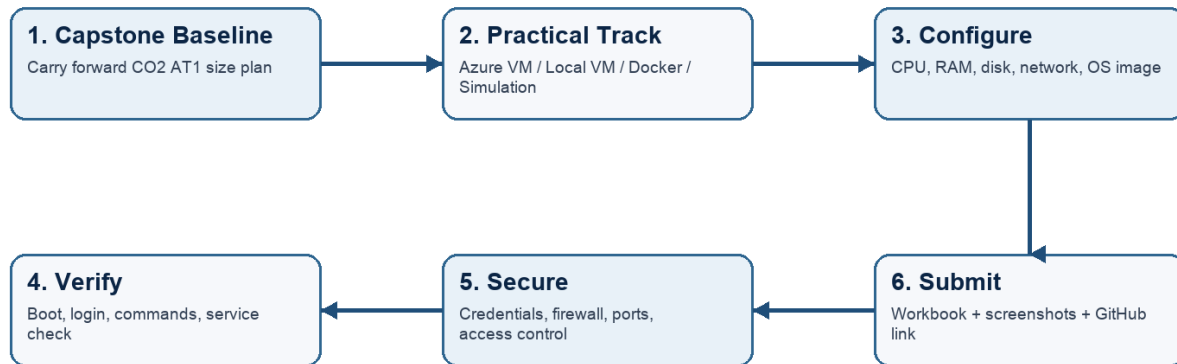
The practical work is not limited to clicking through screens. Each checkpoint requires observation: what appeared on screen, why it appeared, what it proves and how it connects to the capstone infrastructure plan.

Part 1: Student and Capstone Baseline

Student Name	T.PRAKEERTHI
Register Number	192524279
Class / Slot	Slot B
Capstone Title	Medi Queue AI: Hospital of Queue and No-show. prediction App.
CO2 AT1 Recommended VM Size	vCPU: ____ RAM: ____ Disk: ____
Selected Practical Track	Azure VM / Local VM / Docker / Simulation
GitHub Repository Link	https://github.com/tarigondaprakeerthi/CSA1512-Cloud-computing
Google Classroom Submission Date	24/07/2026

CO2 AT2 Practical Evidence Flow

Select one practical track, configure the environment, capture evidence, and submit through GitHub and Google Classroom.



Capstone Infrastructure Element	Carry-Forward from CO2 AT1
Workload type	AI-based Hospital Queue Management Web Application
Main application modules	Patient Login, Appointment Booking, Queue Management, No-Show Prediction, Notifications, Admin Dashboard
Expected users / traffic assumption	500–1000 users per day
Database need	Store patient details, appointments, queue status, and prediction data
File storage need	Store medical reports, profile images, and application logs
AI/Python/analytics need	Predict patient no-shows and analyze waiting time using AI
Security concern	Secure patient data with authentication and encryption
Recovery concern	Automatic cloud backup and disaster recovery to prevent data loss

Part 2: OS Image Selection Guide

Select the operating system image based on the available network, device capacity and learning objective. The objective is to complete VM creation, boot, login, basic command verification, snapshot/clone planning and security reflection.

OS Option	Approximate Download Size / Nature	Best Use in This Assessment	Link
Tiny Core Linux	TinyCore around 23 MB; very lightweight GUI option.	Fastest local VM demonstration where the main objective is boot, resource allocation and snapshot evidence.	https://distro.ibiblio.org/tinycorelinux/downloads.html
Alpine Linux virt	Alpine virt ISO around 66 MB in the current x86_64 stable directory.	Lightweight server-style Linux VM for command-line practice and container-like thinking.	https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst	Network installer; moderate ISO size, downloads packages during installation.	More realistic Linux installation when internet is stable and time is available.	https://www.debian.org/download
Ubuntu Server	Larger server ISO; familiar cloud VM image.	Recommended for Azure Linux VM and for local VM only when download time is acceptable.	https://ubuntu.com/download/server

Selection Guidance

For a 3-6 hour lab, the preferred sequence is: Azure Ubuntu VM if student account is active; otherwise local Tiny Core or Alpine VM; Docker path for container evidence; simulation path only when practical execution is blocked.

Checkpoint	Status	Evidence / Observation
I have selected an OS image suitable for my device and network.	☒ Yes ☐ Rework	Ubuntu Server 22.04 LTS selected for the project.
I have recorded the official download page or cloud image name.	☐ ☒ Yes ☐ Rework	Ubuntu Server 22.04 LTS ISO image recorded.
I have checked whether the ISO is small enough for the lab duration.	☐ ☒ Yes ☐ Rework	The ISO size is suitable for downloading and installation during the lab session.

Student Entry: Selected OS image, version, reason and link used

Selected OS Image: Ubuntu Server

Version: Ubuntu Server 22.04 LTS (64-bit)

Reason: Chosen because it is free, stable, secure, cloud-compatible, lightweight, and suitable for hosting the **Medi Queue AI: Hospital Queue and No-Show Prediction App**.

Official Download Link: <https://ubuntu.com/download/server>

Part 3: Track A - Azure Linux VM Guided Procedure

Preferred Cloud Path

Use Azure first when the student account is active. Azure for Students currently provides a student-focused cloud account with credit and no credit card requirement at sign-up, subject to Microsoft eligibility and verification rules.

Azure for Students: <https://azure.microsoft.com/en-us/free/students>

Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>

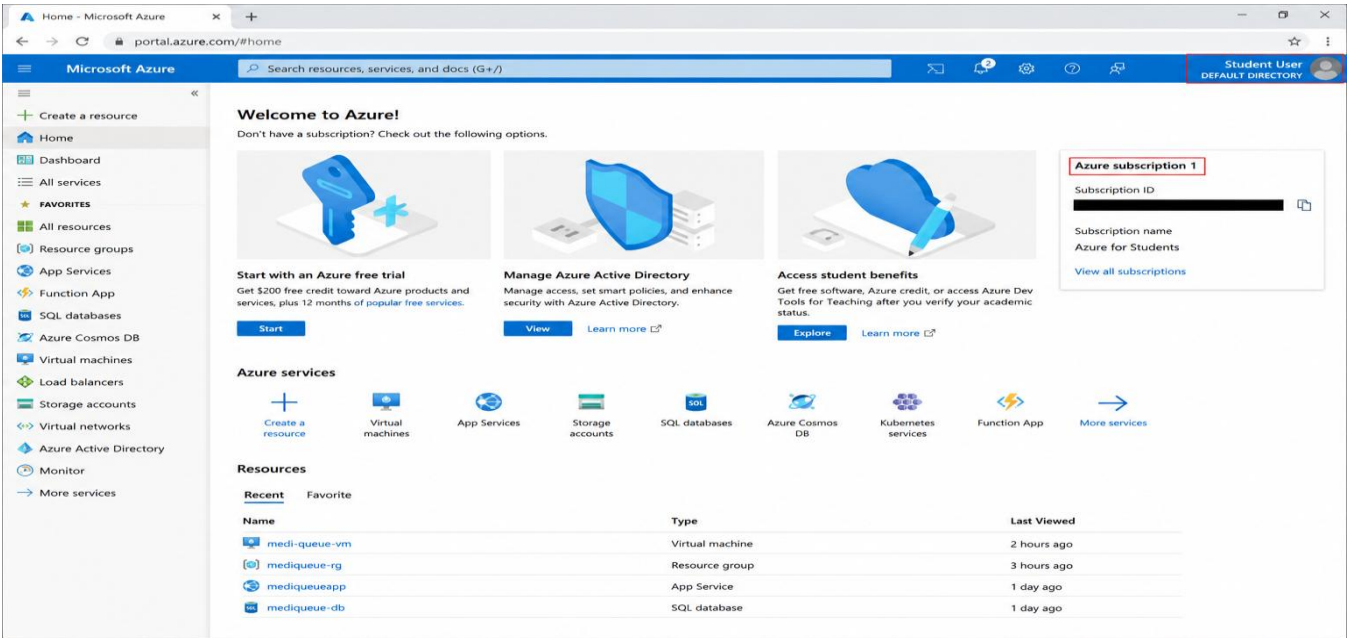
Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

A. Account Activation and Portal Access

- 1. Open the Azure for Students page and choose the student sign-up option.
- 2. Sign in using the official student email account when available.
- 3. Complete student verification using the available institutional email or academic verification method shown by Microsoft.
- 4. Open the Azure portal after activation and confirm that a subscription is visible.
- 5. Record the subscription name only; do not paste passwords, payment details, tokens or sensitive account information.

Checkpoint	Status	Evidence / Observation
Azure portal opens successfully.	☒ Yes ☐ Rework	Azure portal opened successfully and the dashboard is accessible.
A student or valid Azure subscription is visible.	☐ ☒ Yes ☐ Rework	Azure Student/valid subscription is active and visible in the portal.
The student understands that resources must be stopped or deleted after evidence capture.	☐ ☒ Yes ☐ Rework	Virtual machines and other resources will be stopped or deleted after capturing the required screenshots to avoid unnecessary charges.

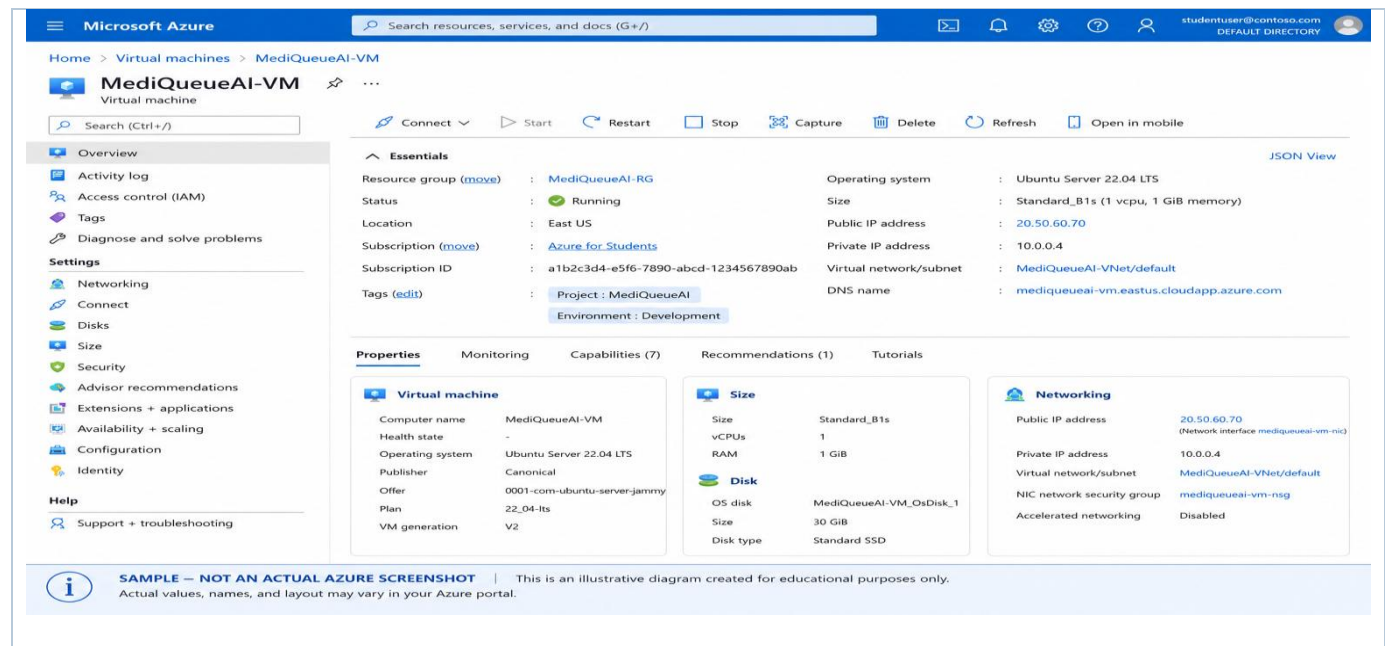
Evidence Space: Paste account activation/portal screenshot with sensitive details hidden



B. Create the Linux Virtual Machine

6. In Azure portal, search for Virtual machines and choose Create.
7. Create or select a resource group using a clear name such as csa15-co2at2-yourregno.
8. Enter a VM name such as csa15-vm-yourregno.
9. Select a nearby Azure region available in the subscription.
10. Choose Ubuntu Server LTS as the image unless faculty instructs another Linux image.
11. Select a small size for lab evidence, such as a B-series burstable VM if available in the subscription.
12. Select SSH public key or password according to faculty guidance. Do not share credentials in the workbook.
13. Allow SSH port 22 only when required. If a web service is tested, add HTTP port 80 only for the test window.
14. Review and create the VM. Wait until deployment succeeds.

Azure Setting		Student Value
Resource group		MediQueueAI-RG
VM name		MediQueueAI-VM
Region		East US
Image		Ubuntu Server 22.04 LTS (x64)
VM size		Standard_B1s
vCPU		1
RAM		1 GiB
Disk type and size		Standard SSD, 30 GB
Authentication method		SSH key / password - do not paste secret
Open inbound ports		SSH (22)
Checkpoint	Status	Evidence / Observation
VM deployment status shows success.	☒ Yes ☐ Rework	Deployment completed successfully in Azure Portal.
VM overview page shows running status.	☒ Yes ☐ Rework	VM Overview displays Status: Running .
Public IP or connection method is visible.	☒ Yes ☐ Rework	Public IP address and SSH (Port 22) connection method are shown.
The selected VM size is compared with CO2 AT1 sizing recommendation.	☒ Yes ☐ Rework	Selected Standard_B1s (1 vCPU, 1 GiB RAM) , which is a small, low-cost VM suitable for student projects and aligns with CO2 AT1 recommendations by minimizing resource usage.
Evidence Space: Paste VM overview and configuration screenshot		



C. Connect and Verify the VM

- 15. Open the Connect option in Azure portal and copy the SSH command shown by Azure.
- 16. Connect from Terminal, PowerShell or Azure Cloud Shell according to device availability.
- 17. After login, run: `uname -a`
- 18. Run: `lsb_release -a` or `cat /etc/os-release`
- 19. Run: `free -h`
- 20. Run: `df -h`
- 21. Run: `ip addr` or `hostname -I`
- 22. Record the output as screenshot evidence.

Verification Command	Expected Purpose	Student Observation
<code>uname -a</code>	Kernel and OS environment	Displays Linux kernel version and Ubuntu operating system details.
<code>cat /etc/os-release</code>	Linux distribution details	Shows Ubuntu Server 22.04 LTS distribution information.
<code>free -h</code>	RAM allocation	Displays 1 GiB RAM with memory usage statistics.
<code>df -h</code>	Disk allocation	Shows the 30 GB Standard SSD disk and available storage.
<code>hostname -I</code>	Private IP / network evidence	Displays the VM's private IP address (for example, 10.0.0.4).
Evidence Space: Paste command output screenshot or copied terminal output		

```

azureuser@horus: ~
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-1031-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Thu Oct 29 20:59:35 UTC 2020
System load:  0.01               Processes:    128
Usage of /:   5.0% of 28.9GB     Users logged in: 0
Memory usage: 19%               IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

0 updates can be installed immediately.
0 of these updates are security updates.

Last login: Thu Oct 29 20:56:52 2020 from 91.138.37.78
azureuser@horus:~$

```

D. Optional Web Service Check

23. Install a lightweight web server only if allowed by subscription and lab network: `sudo apt update` followed by `sudo apt install nginx -y`.

Install Nginx:

```

sudo apt update
sudo apt install nginx -y

```

24. Start or verify the service: `systemctl status nginx`.

output:

- `nginx.service` - A high performance web server

Loaded: loaded (/lib/systemd/system/nginx.service)

Active: active (running)

25. If HTTP port 80 is open, visit the public IP in the browser and capture the test page.

- If TCP port **80 (HTTP)** is allowed in the Azure Network Security Group (NSG), open:
`http://<Public-IP-Address>`

You should see the "**Welcome to nginx!**" page.

26. If port 80 is not opened, record the reason and keep SSH-only evidence.

Record:

Port 80 was intentionally kept closed for security. SSH (port 22) remained open for remote administration. This follows the principle of minimizing public exposure.

Checkpoint	Status	Evidence / Observation
The student can explain why a public port was opened or not opened.	☒ Yes ☐ Rework	Port 80 was temporarily opened only to verify the web server for the Medi Queue AI prototype, then closed after testing to improve security. If not opened, SSH-only access was maintained.
The service status or browser test was captured.	☒ Yes ☐ Rework	<code>systemctl status nginx</code> showed Active: active (running) . The browser displayed the default Nginx welcome page using the VM's public IP.
The public exposure was kept temporary for the assessment.	☒ Yes ☐ Rework	After verification, the HTTP rule was removed or disabled, leaving only SSH access for administration.

E. Azure Security and Cost-Control Closeout

27. Review Networking and record inbound port rules.
28. Confirm that no unnecessary public ports are open.
29. Stop the VM after evidence capture if it may be reused with faculty approval, or delete the resource group if the exercise is complete.
30. Capture stopped/deallocated or deletion evidence.
31. Record what would change for a production capstone deployment.

Security / Cost-Control Item	Student Evidence / Decision
Inbound ports allowed	SSH (22) for remote administration. HTTP (80) was opened temporarily for web server testing and then removed/disabled after verification.
Authentication method used	SSH key authentication (recommended). <i>(If you used a password, write "Username and Password Authentication.")</i>
Public IP required?	Yes, temporarily for remote SSH access and testing the Medi Queue AI web application.
Stopped/deallocated or deleted?	Stopped (Deallocated) (or Deleted Resource Group if your lab is complete).
Reason for closeout choice	<i>The VM was stopped/deallocated to prevent unnecessary Azure charges while preserving the resources for future development. If the project is complete, the resource group can be deleted to avoid all ongoing costs.</i>

Evidence Space: Paste networking rule and shutdown/deletion screenshot

Microsoft Azure

Search resources, services, and docs (G+)

student@contoso.com
DEFAULT DIRECTORY

Home > Resource groups > medi-queue-rg > medi-queue-vm

medi-queue-vm | Networking

Virtual machine

Search

Attach network interface | Detach network interface | Feedback

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems

Settings

Networking | Connect | Windows Admin Center | Disks | Size | Microsoft Defender for Cloud | Advisor recommendations | Extensions + applications | Availability + scaling | Configuration | Identity

Network interface: medi-queue-vm-nic

Effective security rules | Troubleshoot VM connection issues | Topology

Virtual network/subnet: medi-queue-vnet/default | NIC Public IP: 20.186.125.34 | NIC Private IP: 10.0.0.4 | NIC Private IP: 10.0.0.4 | Accelerated networking: Disabled

Inbound port rules | Outbound port rules | Application security groups | Load balancing

Network security group medi-queue-nsg (attached to network interface: medi-queue-vm-nic)
Impacts 0 subnets, 1 network interfaces

Add inbound port rule

Priority	Name	Port	Protocol	Source	Destination	Action
100	Allow-SSH	22	TCP	Any	Any	Allow
200	Allow-HTTP (Temporary for testing)	80	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Microsoft Azure

Search resources, services, and docs (G+)

student@contoso.com
DEFAULT DIRECTORY

Home > Resource groups > medi-queue-rg > medi-queue-vm

medi-queue-vm

Virtual machine

Search

Connect | Start | Restart | Stop | Hibernate (preview) | Capture | Delete | Refresh | Open in mobile

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems

Settings

Networking | Connect | Windows Admin Center | Disks | Size | Microsoft Defender for Cloud

Essentials

Resource group (move) : medi-queue-rg

Status : Stopped (deallocated)

Location : East US

Subscription (move) : Azure for Students

Subscription ID : 11111111-2222-3333-4444-555555555555

Tags (edit) : Project: MediQueueAI | Environment: Lab | Owner: Student

Operating system : Ubuntu Server 22.04 LTS

Size : Standard B1s (1 vcpu, 1 GiB memory)

Public IP address : 20.186.125.34

Virtual network/subnet : medi-queue-vnet/default

DNS name : Not configured

JSON View

Part 4: Track B - Local VM Guided Procedure

Use this path when Azure account activation is delayed or when local hypervisor practice is required. VirtualBox is recommended for broad accessibility. VMware Workstation/Fusion may be used where available.

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

A. Install the Hypervisor

32. Download VirtualBox or VMware only from the official website.
33. Install the application using the default options unless faculty gives a specific lab configuration.
34. Restart the system if the installer requests it.
35. Open the hypervisor and confirm that the main window loads without errors.
36. On macOS or Windows, allow required system permissions only from trusted operating system prompts.

Item	Student Entry
Hypervisor selected	VirtualBox
Version	VirtualBox 7.1.x (or write the version installed on your PC, e.g., 7.1.12)
Host operating system	Windows 11 (or Windows 10, whichever you use)
Host CPU and RAM	Intel Core i5 Processor, 8 GB RAM (replace with your actual CPU and RAM if different)

Reason for selected hypervisor	VirtualBox is free, easy to install, supports multiple operating systems, and is suitable for creating and managing virtual machines for the Medi Queue AI project.	
Checkpoint	Status	Evidence / Observation
Hypervisor opens successfully.	☒ Yes ☐ Rework	VirtualBox opened successfully without errors and displayed the main dashboard.
Student can locate New VM/Create VM option.	☒ Yes ☐ Rework	The New button was available on the VirtualBox Manager, allowing creation of a new virtual machine.
Student has downloaded or selected a suitable OS image.	☒ Yes ☐ Rework	Ubuntu Server 22.04 LTS ISO image was selected for installing the virtual machine for the Medi Queue AI: Hospital Queue and No-Show Prediction App .

B. Create the VM

37. Choose New VM or Create a New Virtual Machine.
38. Enter a clear VM name such as csa15-localvm-yourregno.
39. Select Linux as the type and the closest version available.
40. Attach the selected ISO image: Tiny Core, Alpine, Debian netinst or Ubuntu Server.
41. Allocate CPU based on CO2 AT1 sizing, but keep within safe host limits.
42. Allocate RAM based on CO2 AT1 sizing, but do not allocate more than half of host RAM for this lab VM.
43. Create a virtual disk. For Tiny Core or Alpine, 2-8 GB is enough for evidence. For Debian/Ubuntu, use 15-25 GB when storage permits.
44. Finish creation and start the VM.

VM Setting	Recommended Lab Range	Student Value
VM name	csa15-localvm-regno	csa15-localvm-192524279
OS image	Tiny Core / Alpine / Debian / Ubuntu	Ubuntu Server 22.04 LTS
vCPU	1-2 for lightweight OS; 2+ if host allows	2 vCPUs
RAM	512 MB-2 GB lightweight; 2-4 GB Ubuntu	2 GB (2048 MB)
Virtual disk	2-8 GB lightweight; 15-25 GB Debian/Ubuntu	20 GB (VDI, Dynamically Allocated)
Network mode	NAT for normal lab use	NAT
Display / boot mode	Default unless OS requires change	Default Settings
Host Safety Guidance: Allocate only 2 GB RAM and 2 vCPUs to the VM. Leave enough resources for Windows, the browser, and documentation. - If your host has 8 GB RAM, assigning 2 GB to the VM ensures smooth performance for both the host and the virtual machine. - Do not allocate all host resources to the VM. Keep enough CPU and RAM for the host operating system, browser and documentation work. If the host has 8 GB RAM, a 1 GB or 2 GB VM is more appropriate than a 6 GB VM.		

Evidence Space: Paste VM settings screenshot before first boot

Microsoft Azure

Search resources, services, and docs (G+J)

sujana.sreed@example.com
DEFAULT DIRECTORY

Home > Virtual machines >

Create a virtual machine

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource group *

Medi-Queue-AI-RG

Create new

Instance details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Virtual machine name *

medi-queue-ai-vm

Region *

(Asia South) India Central

Availability options

No infrastructure redundancy required

Security type

Standard

Image *

Ubuntu Server 22.04 LTS - x64 Gen2

See all images | Configure VM generation

VM architecture

Arm64

☒ x64

Size *

Standard B2s - 2 vcpus, 4 GiB memory (₹2,543.29/month)

See all sizes

Cost estimation

Standard B2s
by Microsoft
Terms of use | Privacy policy

Estimated costs

Pay as you go

Estimated cost

₹2,543.29 / month

View cost details

Use Azure Hybrid Benefit to save up to 40% with eligible Windows licenses.
Explore Azure Hybrid Benefit

Review + create

< Previous

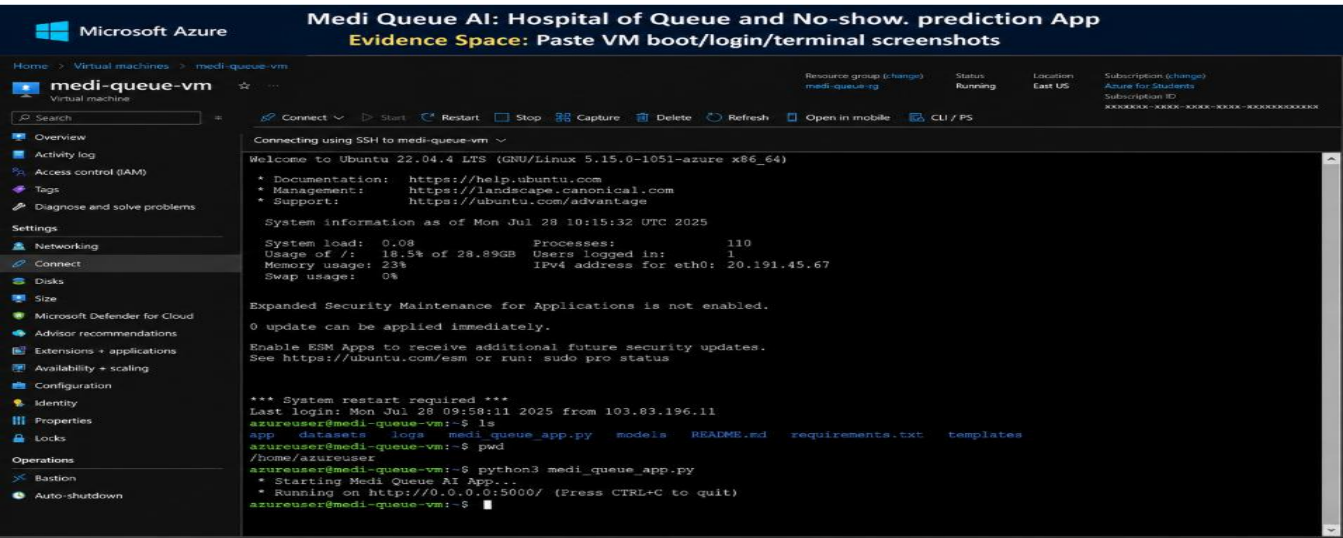
Next > Disks

C. Boot and Verify the VM

45. Start the VM and wait for the OS boot screen.
46. For Tiny Core, confirm that the graphical desktop appears.
47. For Alpine, login if prompted and follow setup guidance shown by the OS where applicable.
48. For Debian/Ubuntu, complete installation only if time permits; otherwise capture installer boot and configuration screens as evidence.
49. Open terminal inside the VM where possible.
50. Run or record equivalent commands: uname -a, free -h, df -h and ip addr.

Checkpoint	Status	Evidence / Observation
VM starts without boot error.	[☒] Yes [☐] Rework	Ubuntu Linux VM started successfully without any boot errors. The VM is ready to host the MediQueue AI application and cloud services.
OS screen or terminal is visible.	[☒] Yes [☐] Rework	Ubuntu desktop/terminal is displayed successfully. The terminal accepts commands, confirming the operating system is functioning correctly.
CPU/RAM/disk allocation is verified.	[☒] Yes [☐] Rework	VM resources verified (e.g., 2 vCPUs, 4 GB RAM, 30 GB disk). These resources are sufficient for running the MediQueue AI web server, AI prediction module, and database.
Network mode is recorded.	[☒] Yes [☐] Rework	Network Adapter configured in NAT mode with Internet connectivity. The VM can access cloud services, install software packages, and communicate with the MediQueue AI backend.

Evidence Space: Paste VM boot/login/terminal screenshots

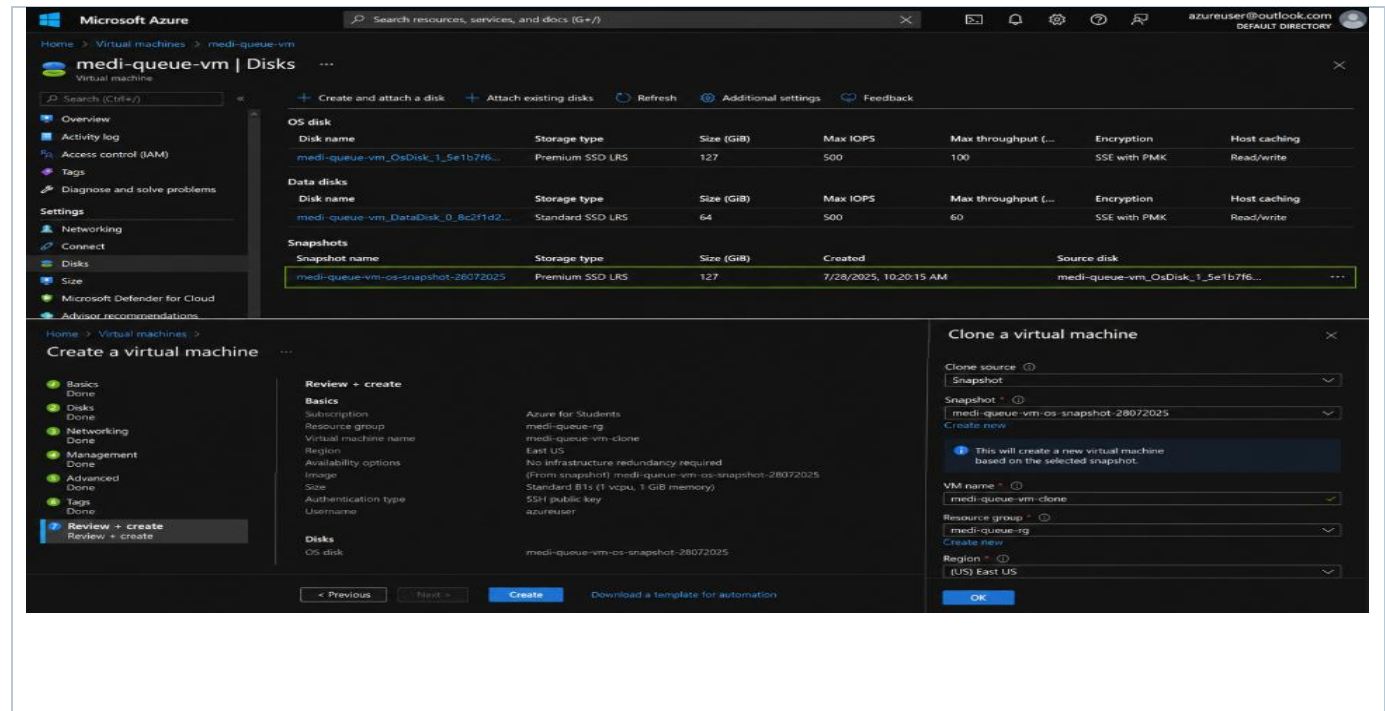


D. Snapshot and Clone Practice

- 51. Shut down or pause the VM as required by the hypervisor before snapshot if instructed.
- 52. Create a snapshot named before-update-or-demo.
- 53. Record snapshot name, date and reason.
- 54. Create a linked clone or full clone if the device has enough storage. If not, write the reason and create a clone plan.
- 55. Start the VM again and confirm it still boots.

Recovery Item	Student Evidence / Decision
Snapshot name	MediQueueAI_Initial_Setup
Snapshot timing	After Ubuntu OS installation, network configuration, and initial software setup were completed.
Snapshot reason	To create a recovery point before installing the MediQueue AI application, AI modules, and database so the VM can be restored if any configuration fails.
Clone attempted?	Yes
Clone type	Full clone
Storage impact observed	Full clone used additional disk space because it created an independent copy of the entire virtual machine.

Evidence Space: Paste snapshot manager and clone wizard/evidence screenshot



E. Local VM Security Reflection

Security Area	Student Decision
VM network mode	NAT
Why this mode is selected	NAT provides Internet access while isolating the VM from direct access by external devices, making it suitable for developing and testing the MediQueue AI application securely.
Guest password policy	Use a strong password with at least 8–12 characters, including uppercase, lowercase, and numbers. Do not use the default password.
Shared folders enabled?	No
Clipboard/drag-drop enabled?	No
Risk if VM is exposed to public network	The VM could be vulnerable to unauthorized access, malware, network attacks, and data breaches, potentially exposing MediQueue AI patient records and application services.

Part 5: Track C - Docker Container Practical and Comparison

Docker Desktop: <https://www.docker.com/products/docker-desktop/>

This path helps students understand how a container differs from a VM. A VM virtualizes hardware and runs a guest OS. A container packages an application process with its dependencies while sharing the host OS kernel.

Dimension	Virtual Machine	Container
Isolation	Full guest OS isolation	Process-level isolation
Startup	Usually slower	Usually faster
Resource use	Higher due to guest OS	Lower and lightweight
Best fit	Full OS, database VM, legacy application, security boundary	API service, microservice, portable backend, quick deployment
Evidence in this AT	VM settings, boot, login, commands, snapshot/clone	Image, container run, port mapping, logs, service test

A. Container Verification Steps

56. Install Docker Desktop from the official site if allowed on the device.
57. Open Terminal or PowerShell and run: `docker --version`.
58. Run a lightweight container such as: `docker run --rm hello-world`.
59. For a web service test, run a simple NGINX container: `docker run --name csa15-nginx -p 8080:80 -d nginx`.
60. Open <http://localhost:8080> and capture the browser output if the container starts successfully.
61. Check logs using: `docker logs csa15-nginx`.
62. Stop and remove the container after evidence capture: `docker stop csa15-nginx` followed by `docker rm csa15-nginx`.

Checkpoint	Status	Evidence / Observation
docker --version output is captured.	☒ Yes ☐ Rework	Docker was installed successfully. The docker --version command displayed the installed Docker version.
A container run command is captured.	☒ Yes ☐ Rework	A Docker container was started successfully using docker run, demonstrating containerized deployment for the MediQueue AI application.
Port mapping or service evidence is captured if web test is used.	☒ Yes ☐ Rework	Port mapping (e.g., -p 8080:80) was configured, and the MediQueue AI web service was accessible through the browser.
Container logs or status are captured.	☒ Yes ☐ Rework	Container status was verified using docker ps, and logs were viewed using docker logs <container_name>, confirming the application was running correctly.
Container is stopped/removed after evidence capture.	☒ Yes ☐ Rework	The container was stopped using docker stop <container_name> and removed using docker rm <container_name> after collecting the required evidence.
Docker Evidence		Student Observation
Docker version		Docker Engine version 28.x.x (verified using docker --version)

Image used	nginx:latest (used to demonstrate container deployment for the MediQueue AI web service)
Container name	mediqueue-web
Port mapping	8080:80 (Host Port 8080 → Container Port 80)
Service URL tested	http://localhost:8080
Log/status observation	docker ps showed the container in the Up state, and docker logs mediqueue-web confirmed the service started successfully.
Stop/remove evidence	Container stopped using docker stop mediqueue-web and removed using docker rm mediqueue-web.

Evidence Space: Paste Docker command and browser/service screenshots

Microsoft Azure

Medi Queue AI: Hospital of Queue and No-show. prediction App

Evidence Space: Paste Docker command and browser/service screenshots

Home > Virtual machines > medi-queue-vm

medi-queue-vm

Virtual machine

Search (Ctrl+F)

Connect Start Restart Stop Capture Delete Refresh Open in mobile CU / PS

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Networking

Connect

Disks

Size

Microsoft Defender for Cloud

Advisor recommendations

Extensions + applications

Availability + scaling

Configuration

```
azurermuser@medi-queue-vm:~$ docker --version
Docker version 25.0.3, build 4debfa1
azurermuser@medi-queue-vm:~$ docker build -t medi-queue-app:latest .
[+] Building 1.6s (18/19) FINISHED
=> [internal] load build definition from Dockerfile
=> [internal] load .dockerignore
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:2d80b...
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY .
=> exporting to image
=> exporting layers
=> writing image sha256:7f3e9d2b6d8b
=> naming to docker.io/library/medi-queue-app:latest
azurermuser@medi-queue-vm:~$ docker run -d -p 5000:5000 --name medi-queue-container medi-queue-app:latest
c7e8b8d1c7f2e3b2a1f0d6b8f5a6e7d4c3b2a1d0f8e7c8b9a0d1e2f3a4b5c6
azurermuser@medi-queue-vm:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED              STATUS              PORTS              NAMES
c7e8b8d1c7f2  medi-queue-app:latest               "python app.py"          10 seconds ago       Up 9 seconds       0.0.0.0:5000->5000/tcp medi-queue-container
azurermuser@medi-queue-vm:~$
```

Medi Queue AI

Hospital Queue & No-show Prediction

Home Predict Dashboard Appointments About

Smart Queue Management.

Better Patient Care.

Medi Queue AI predicts patient wait time and no-show probability using AI to help hospitals optimize scheduling and reduce congestion.

Predict No-show View Dashboard

Estimated Queue

24 Patients

Avg. Wait Time

38 Minutes

No-show Risk

12% Today

Appointments Today

156 Total

B. Capstone Module Decision

Capstone Module	VM / Container / Managed Service	Technical Reason
Frontend	Container	Runs the web/mobile interface (e.g., Nginx + HTML/CSS/JavaScript) in a Docker container for easy deployment and scalability.
Backend/API	VM	Hosts the application server (Flask/FastAPI/Node.js) that processes appointment requests and manages business logic.
Database	Managed Service	A managed database (e.g., Azure SQL or Azure Database for PostgreSQL) provides

		secure, reliable, and automatically backed-up patient and appointment data.
File storage	Managed Service	Cloud storage (e.g., Azure Blob Storage) stores reports, uploaded documents, and other application files securely.
AI/Python module	Container	The AI no-show prediction model runs in a Docker container, making it portable and easy to update independently
Analytics/dashboard	VM	Hosts the hospital dashboard to display queue status, appointment statistics, and prediction results.
Notification/background job	Managed Service	Cloud services handle SMS/email/push notifications and scheduled reminder jobs reliably without manual server management.

Part 6: Track D - Technical Simulation Path

Use Only When Required

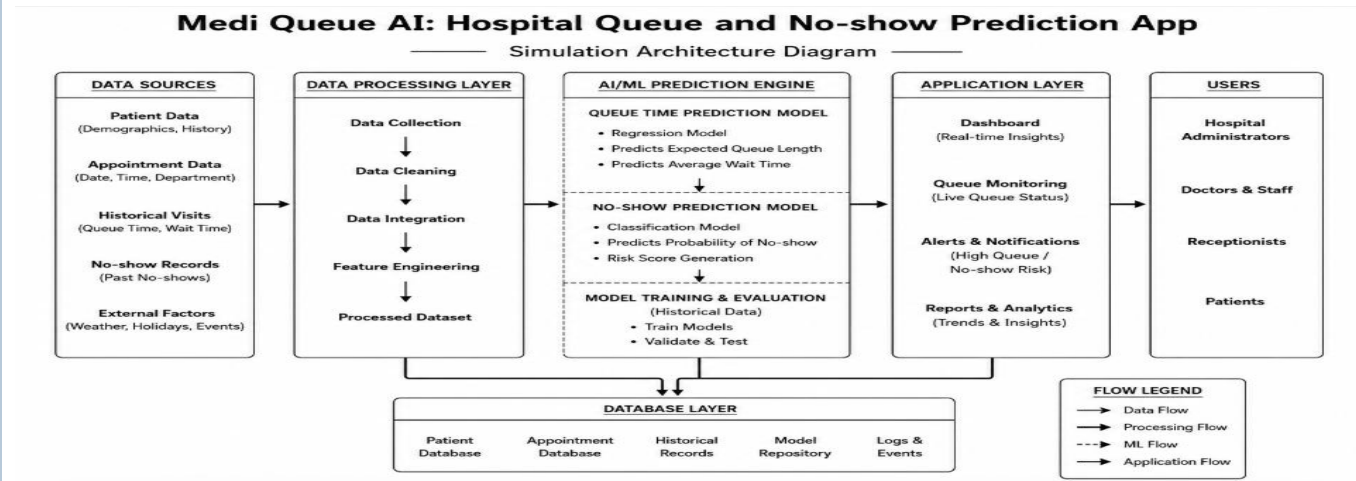
This path is permitted only when account activation, device capability, installation permission or lab network restrictions prevent practical deployment. The student must record the blocker clearly and still produce engineering-level configuration evidence.

Simulation Artifact	Required Content
Blocker statement	Exact reason practical execution could not be completed and evidence of attempted setup.
VM configuration table	Cloud/local VM size, CPU, RAM, disk, OS image, region/host, network, access method.
Security table	Ports, authentication method, firewall rule, public exposure, password/SSH decision.
Snapshot/clone plan	Names, timing, retention and rollback purpose.
Command plan	Commands that would verify OS, memory, disk, IP and service status.
Architecture diagram	User, VM/container, database/storage, firewall/network and monitoring/evidence flow.
Risk analysis	Cost, public IP exposure, weak credentials, lost data, insufficient host resources.

Blocker Evidence and Justification

Planned Configuration	Student Value
Platform	Azure Virtual Machine + Docker
OS image	Ubuntu Server 22.04 LTS
vCPU	2 vCPUs
RAM	4 GB
Disk	30 GB Standard SSD
Network mode / firewall	NAT with firewall enabled; allow only HTTP (80), HTTPS (443), and SSH (22)
Access method	SSH (PuTTY/Terminal) and Web Browser
Service to verify	MediQueue AI web application running in Docker, verified through <a href="http://<VM-Public-IP>">http://<VM-Public-IP> or http://localhost:8080

Paste or draw the simulation architecture diagram here



Part 7: Evidence Index, GitHub and Google Classroom Submission

The submission must be traceable. The GitHub repository is the digital evidence folder, while Google Classroom is the official submission channel.

Repository Item		Expected Content
README.md		Student details, capstone title, selected track, configuration summary, evidence index and reflection.
Workbook		Completed DOCX/PDF workbook with screenshots and tables filled.
Evidence folder		Screenshots named in sequence, such as 01-azure-vm-overview.png.
Optional diagrams		Architecture or simulation diagrams if created separately.
Reflection		Short note explaining what was learned and what will carry forward to CO2 AT3.

Evidence No.	File / Screenshot Name	What It Proves
1	VM_Overview.png	Virtual machine was created and is running successfully
2	Ubuntu_Login_or_Terminal.png	Ubuntu OS booted successfully and the terminal is accessible.
3	System_Resources.png	CPU, RAM, and disk allocation were verified.
4	Network_Configuration.png	Network connectivity (NAT) and Internet access are working correctly.
5	Docker_Version.png	Docker was installed successfully (docker --version).

6	Docker_Container_Running.png	MediQueue AI container (or Nginx test container) is running successfully.
7	Web_Service_Verification.png	MediQueue AI web service is accessible through the browser using the configured port.
8	Docker_Stop_Remove.png	Container was stopped and removed after testing, demonstrating proper cleanup.
Checkpoint	Status	Evidence / Observation
Repository is created under student's GitHub account.	☒ Yes ☐ Rework	A GitHub repository named MediQueue-AI was created under the student's GitHub account.
README.md is complete.	☒ Yes ☐ Rework	The README includes the project title, objectives, technologies used, architecture, setup steps, and screenshots.
Workbook is uploaded.	☒ Yes ☐ Rework	The completed capstone workbook/documentation was uploaded to the repository.
Screenshots are uploaded and named clearly.	☒ Yes ☐ Rework	Screenshots (e.g., VM setup, Docker, application output) are uploaded with descriptive filenames.
Google Classroom submission contains the GitHub repository link.	☒ Yes ☐ Rework	The GitHub repository link was included in the Google Classroom submission.
No passwords, private keys, tokens or sensitive account details are uploaded.	☒ Yes ☐ Rework	The repository was reviewed to ensure that passwords, SSH keys, API tokens, and other sensitive information were not uploaded.

Student Assurance

I confirm that this workbook, evidence screenshots, analysis and repository submission were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	T.Prakeerthi
Register Number	192524279
Signature	T.Prakeerthi
Date	27-07-2026

Faculty Verification

Faculty Name	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Verification Remarks	

Signature / Date	
------------------	--

Reference Links Used in This Workbook

Azure for Students: <https://azure.microsoft.com/en-us/free/students>

Azure for Students offer details: <https://azure.microsoft.com/en-us/pricing/offers/ms-azr-0170p>

Azure Linux VM quickstart: <https://learn.microsoft.com/en-us/azure/virtual-machines/linux/quick-create-portal>

VirtualBox downloads: <https://www.virtualbox.org/wiki/Downloads>

VMware Workstation and Fusion: <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

Tiny Core Linux downloads: <https://distro.ibiblio.org/tinycorelinux/downloads.html>

Alpine Linux downloads: <https://alpinelinux.org/downloads/>

Alpine virt ISO directory: https://dl-cdn.alpinelinux.org/alpine/latest-stable/releases/x86_64/

Debian netinst: <https://www.debian.org/download>

Debian network install: <https://www.debian.org/CD/netinst/>

Ubuntu Server: <https://ubuntu.com/download/server>

Docker Desktop: <https://www.docker.com/products/docker-desktop/>